

ALGEBRA

- $(a+b)^2 = a^2 + 2ab + b^2$.
- $(a-b)^2 = a^2 - 2ab + b^2$.
- $(a+b)^2 = (a-b)^2 + 4ab$.
- $(a-b)^2 = (a+b)^2 - 4ab$.
- $(a+b)^2 + (a-b)^2 = 2(a^2 + b^2)$.
- $(a+b)^2 - (a-b)^2 = 4ab$.
- $ab = \left(\frac{a+b}{2}\right)^2 - \left(\frac{a-b}{2}\right)^2$.
- $a^2 + b^2 = (a+b)^2 - 2ab = (a-b)^2 + 2ab$.
- $a^2 - b^2 = (a+b)(a-b)$.
- $(x+a)(x+b) = x^2 + x(a+b) + ab$.
- $(a+b+c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$
- $a^2 + b^2 + c^2 = (a+b+c)^2 - 2(ab+bc+ca)$.
- $2(ab+bc+ca) = (a+b+c)^2 - (a^2 + b^2 + c^2)$.
- $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3 = a^3 + b^3 + 3ab(a+b)$.
- $(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3 = a^3 - b^3 - 3ab(a-b)$
- $a^3 + b^3 = (a+b)^3 - 3ab(a+b)$.
- $a^3 + b^3 = (a+b)(a^2 - ab + b^2)$.
- $a^3 - b^3 = (a-b)^3 + 3ab(a-b)$.
- $a^3 - b^3 = (a-b)(a^2 + ab + b^2)$.
- $(a+b)^3 + (a-b)^3 = 2a^3 + 6ab^2 = 2a(a^2 + 3b^2)$.
- $(a+b)^3 - (a-b)^3 = 6a^2b + 2b^3 = 2b(3a^2 + b^2)$.
- $a^3 + b^3 + c^3 - 3abc = (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca)$.
- $a^3 + b^3 + c^3 - 3abc = \frac{1}{2}(a+b+c)\{(a-b)^2 + (b-c)^2 + (c-a)^2\}$.
- If $a+b+c=0$, then $a^3 + b^3 + c^3 = 3abc$.
- $a^4 + a^2b^2 + b^4 = (a^2 + ab + b^2)(a^2 - ab + b^2)$.
